

Multiple Choice (10 marks)

1. Which of the following statements is FALSE about memory reclamation based on reference counting?
 - (a) Reference counting is well suited for reclaiming cyclic structures.
 - (b) Reference counting incurs additional space overhead for each memory cell.
 - (c) Reference counting is an alternative to mark-and-sweep garbage collection.
 - (d) Reference counting need not keep track of which cells point to other cells.
2. Of the following, which gives the best upper bound for the value of $f(N)$ where f is a solution to the recurrence $f(2N + 1) = f(2N) = f(N) + \log N$ for $N \geq 1$, with $f(1) = 0$?
 - (a) $O(\log N)$
 - (b) $O(N \log N)$
 - (c) $O(\log N) + O(1)$
 - (d) $O((\log N)^2)$
3. In Python 3, which of the following function sets the integer starting value used in generating random numbers?
 - (a) `choice(seq)`
 - (b) `randrange ([start,] stop [,step])`
 - (c) `random()`
 - (d) `seed([x])`
4. Which of the following is not a sentence that is generated by the grammar $A \rightarrow BC$, $B \rightarrow x|Bx$, $C \rightarrow B|D$, $D \rightarrow y|Ey$, $E \rightarrow z$?
 - (a) `xyz`
 - (b) `xy`
 - (c) `xxzy`
 - (d) `xxxxy`
5. Let $x = 8$. What is $x >> 1$ in Python 3?
 - (a) 3
 - (b) 4
 - (c) 2
 - (d) 8

True / False (10 marks)

Answer True or False

6. True or False: The function $O(N^{1/N})$ grows slower than $O(N^{1/4})$, making it the slowest growing function among these options.
7. True or False: The boolean expression $a[i] == \max \parallel !(\max != a[i])$ simplifies to $a[i] != \max$.
8. True or False: Authentication is compromised when an unauthorized individual uses stolen credentials to access a system.
9. True or False: An Internet Protocol (IP) address is assigned to a device when it connects to the Internet.
10. True or False: Having several different classes (sets) of registers is not generally considered an obstacle to aggressive pipelining of an integer unit.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function that matches a string that has an a followed by zero or more b's by using regex.
12. Write a function to find the surface area of a cube.

End of Paper